

Engineering Physics: BAPHY105

Postulates of Quantum Mechanics - I

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Lecture Outcomes

- 1 Basic Postulates of Quantum Mechanics
- 2 Postulate 1: State of a System

Postulates of Quantum Mechanics

- The formalism of quantum mechanics is based on **five postulates** which are the outcome of a **range of experimental observations**.
- The prediction of these postulates has been verified by a rich collection of experiments.
- In **classical mechanics**, the state of a particle at any time t is specified by two fundamental dynamic variables: the **position $\vec{q}(t)$** and the **momentum $\vec{p}(t)$** . All other physical quantity relevant to the particle can be calculated in terms of these two dynamic variables.
- The quantum mechanical counterparts of these ideas is captured in terms of specific postulates which enable us to understand:
 - ① How a quantum state is described mathematically at a given time t ?
 - ② How to calculate the various physical quantities from this quantum state?
 - ③ How to describe the time evolution of a quantum state corresponding to a system?

Postulate 1: State of a System

- ✓ The state of any physical system at time t is specified by a state vector $|\psi(t)\rangle$ in a Hilbert space $\mathcal{H}$.
- ✓ $|\psi(t)\rangle$ contains all the needed information about the system.
- ✓ Any superposition of state vectors is also a state vector.
- ✓ The abstract state vector $|\psi(t)\rangle$ can be represented in a basis as:
 - ① Position basis $\rightarrow$ Position representation: $\psi(x, t) = \langle x | \psi(t) \rangle$
 - ② Momentum basis $\rightarrow$ Momentum representation: $\phi(p, t) = \langle p | \psi(t) \rangle$
- ✓ The wave functions must be physically acceptable:
 - The wave functions along with their first derivatives must be **continuous, single values and finite** everywhere.
 - The wave function should also be **square integrable**.

- ✓ According to Born's probabilistic interpretation, the square of the norm of $\psi(x, t)$ represents the position probability density.

$$P(x, t) = |\psi(x, t)|^2$$

- ✓ Similarly, the square of the norm of $\phi(p, t)$ represents the momentum probability density.

$$P(p, t) = |\phi(p, t)|^2$$

- ✓ The state of a system does not have to be represented by a single wave function. It can be represented by a **superposition of two or more wave functions**.

$$|\psi\rangle = \sum_i \alpha_i |\psi_i\rangle$$

- ✓ Here, α_i are complex numbers and the quantity

$$P = \left| \sum_i \alpha_i |\psi_i\rangle \right|^2 = |\alpha_i|^2 = P_1 + P_2 + P_3 + \dots = 1$$

represents the probability for this superposition if $|\psi_i\rangle$ are normalized.

- ✓ Here, P_i is the probability of finding the system in the state $|\psi_i\rangle$.

Problem Consider a system whose state is given in terms of an orthonormal set of three vectors: $|\phi_1\rangle$, $|\phi_2\rangle$ and $|\phi_3\rangle$ as,

$$|\psi\rangle = \frac{1}{\sqrt{3}}|\phi_1\rangle + \frac{2}{3}|\phi_2\rangle + \frac{\sqrt{2}}{3}|\phi_3\rangle$$

verify that $|\psi\rangle$ is normalized. Calculate the respective probabilities of finding the system in any one of the states $|\phi_1\rangle$, $|\phi_2\rangle$ and $|\phi_3\rangle$. verify that the total probability is one.

Solution:

$$\langle\psi|\psi\rangle = \frac{1}{3}\langle\phi_1|\phi_1\rangle + \frac{4}{9}\langle\phi_2|\phi_2\rangle + \frac{2}{9}\langle\phi_3|\phi_3\rangle$$

$$\Rightarrow \langle\psi|\psi\rangle = \frac{1}{3} + \frac{4}{9} + \frac{2}{9} = 1$$

Therefore $|\psi\rangle$ is normalized.

Since, $|\psi\rangle$ is normalized, the probability of finding the system in $|\phi_1\rangle$ is,

$$P_1 = |\langle\phi_1|\psi\rangle|^2 = \left| \frac{1}{\sqrt{3}}\langle\phi_1|\phi_1\rangle + \frac{2}{3}\langle\phi_2|\phi_2\rangle + \frac{\sqrt{2}}{3}\langle\phi_3|\phi_3\rangle \right|^2 = \frac{1}{3}$$

Similarly,

$$P_2 = \frac{4}{9}; \quad P_3 = \frac{2}{9}$$

$$\therefore P = P_1 + P_2 + P_3 = \frac{1}{3} + \frac{4}{9} + \frac{2}{9} = 1$$